

Variational JEPA – Adding Uncertainty to Visual Representations

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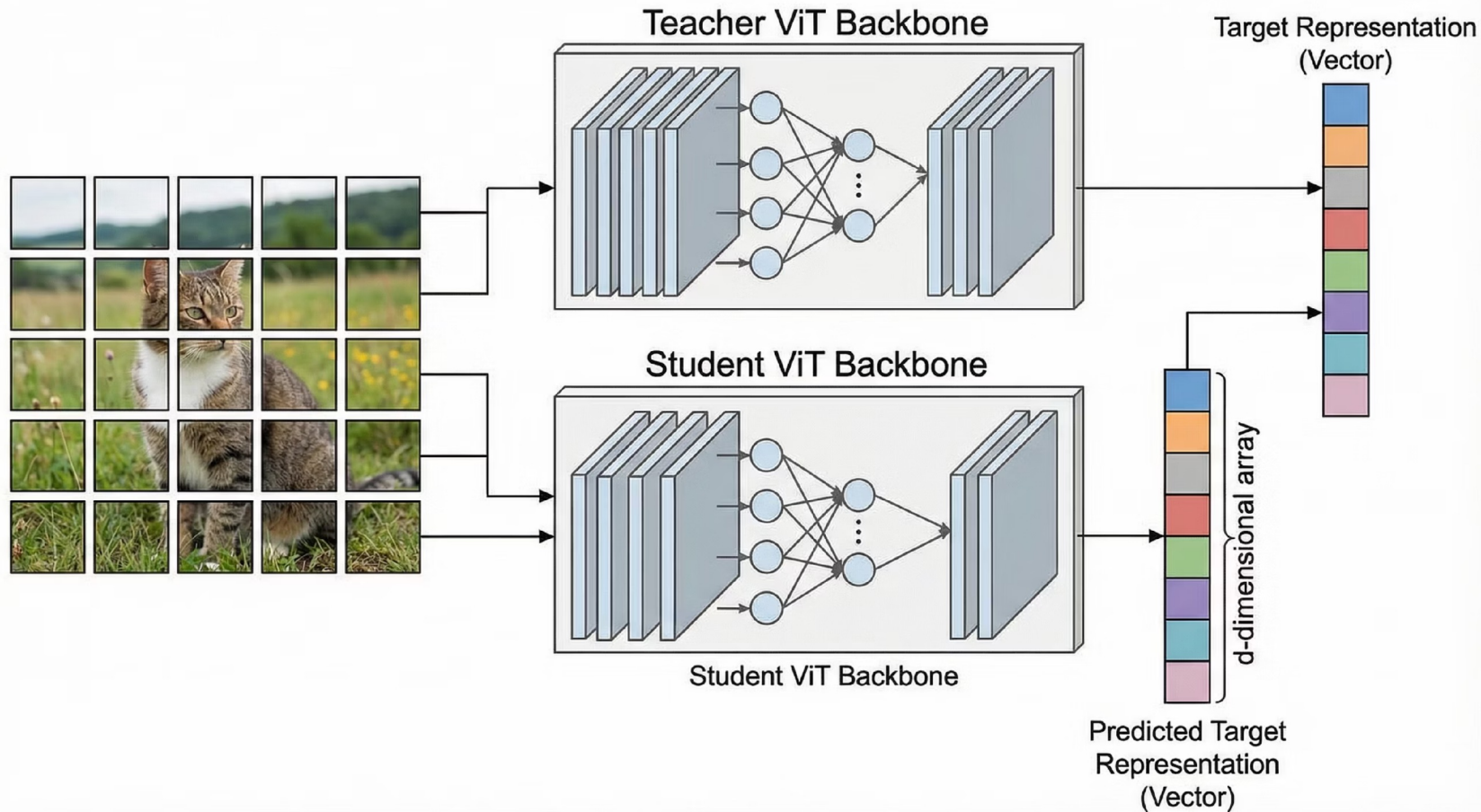
Research Overview

The Challenge

Standard I-JEPA predictors ignore uncertainty and deterministically output a single embedding, despite the missing part possibly having multiple interpretations

Our Solution

Four probabilistic predictors that model latent distributions for masked image blocks, capturing uncertainty while maintaining semantic representations



Why Probabilistic Predictors Matter



Multimodal Completions

Scenes support multiple plausible completions — occluded limbs, textured backgrounds, ambiguous categories



Epistemic Uncertainty

Single deterministic output hides uncertainty and often leads to overconfident predictions



Calibrated Predictions

Sampling from learned distributions enables Monte Carlo estimates and active data selection

Joint-Embedding Predictive Architecture (JEPA)

JEPAAs train a model to **predict the teacher's latent embeddings** for masked image regions using only the **visible context**.

Instead of reconstructing pixels, they operate entirely in **representation space**, which encourages learning **high-level semantic features** rather than low-level details.

Core JEPA Pipeline



Mask Sampling

select context blocks and masked target blocks from the image



Encoding

the teacher encoder sees the full image; the student sees only the context



Prediction

Predictor outputs distribution for each masked target token



Optimization

Loss applied in latent space, EMA updates teacher

Four Probabilistic Approaches

1

Mixture Density Network (MDN)

Multivariate mixture of Gaussians for simple multimodal modeling

2

Autoregressive RNADE

Masked MLP with factorized density estimation

3

RealNVP Flow

Normalizing flow with exact likelihoods via flow matching

4

Diffusion Head

Iterative denoising of latent embeddings

Experimental Setup

Dataset

Tiny ImageNet-100

- 100 classes
- 50k train / 5k validation
- 64×64 resolution

Architecture

ViT Encoder

- Patch size: 8×8
- 10 transformer layers
- Hidden dim: 384
- 8 attention heads

Training Details

- AdamW optimizer with gradient clipping ($\|g\|_2 \leq 3$)
- EMA momentum: 0.998 $\rightarrow$ 1.0 (cosine schedule)
- Random resized crop (80-100% area) + horizontal flip
- Target blocks: area $\in [0.15, 0.20]$, aspect ratio $\in [0.75, 1.50]$
- Minimum 8 context patches enforced

Approach 1: Mixture Density Network

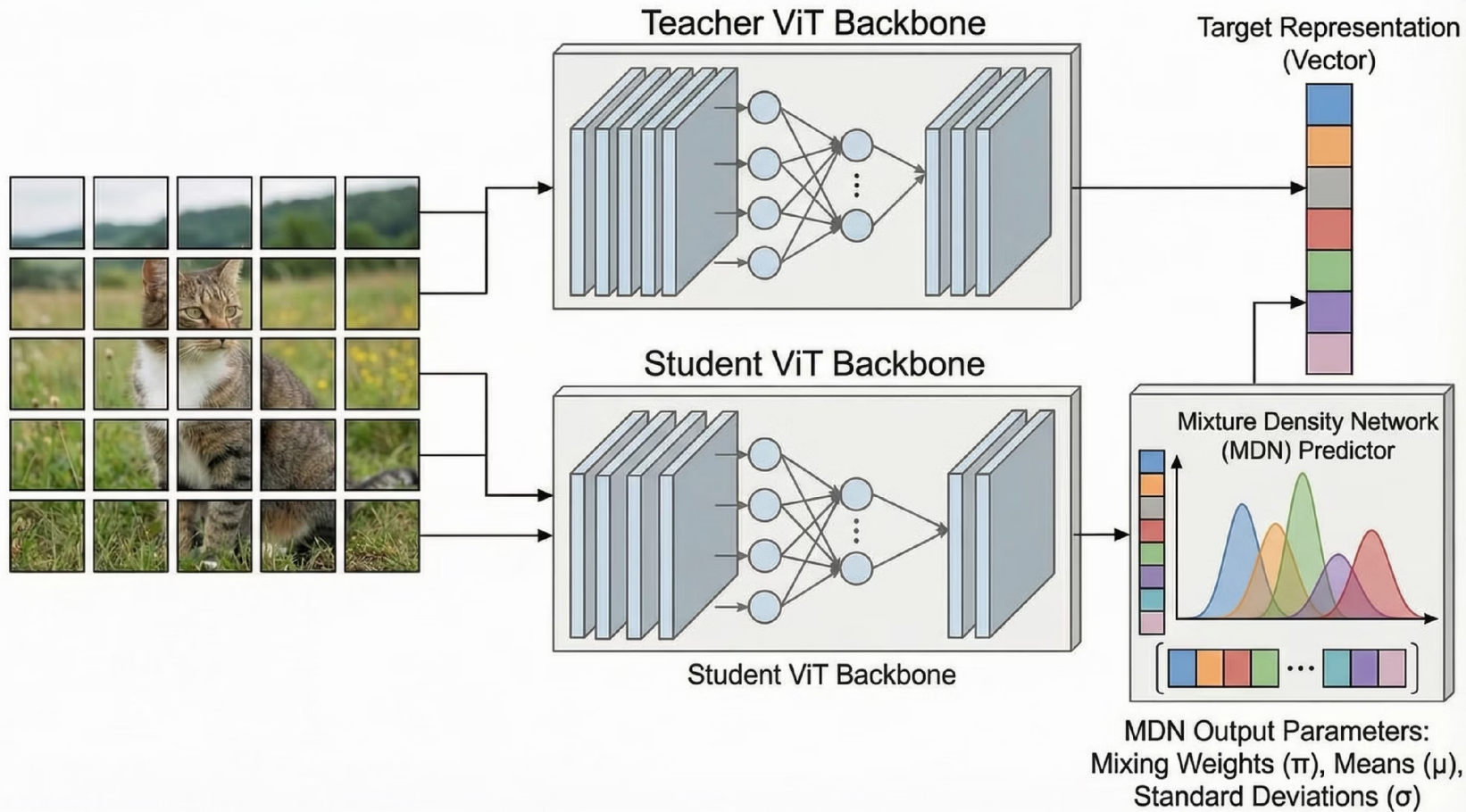
Architecture

e

Two transformer layers process context and target queries, then three linear heads emit mixture weights (π_k), means (μ_k), and variances (σ_k) for $K=5$ diagonal Gaussians per token

Loss Function

Negative log-likelihood of teacher tokens under the mixture distribution. Single forward pass provides explicit parametric distribution



Approach 2: Autoregressive RNADE

Factorizes joint density into one-dimensional conditionals ordered along embedding coordinates. Masked MLP with FiLM modulation incorporates pooled context, making every conditional $p(z_a | z_{(a)}, \text{context})$ explicitly context-aware.

01

Normalize embeddings

02

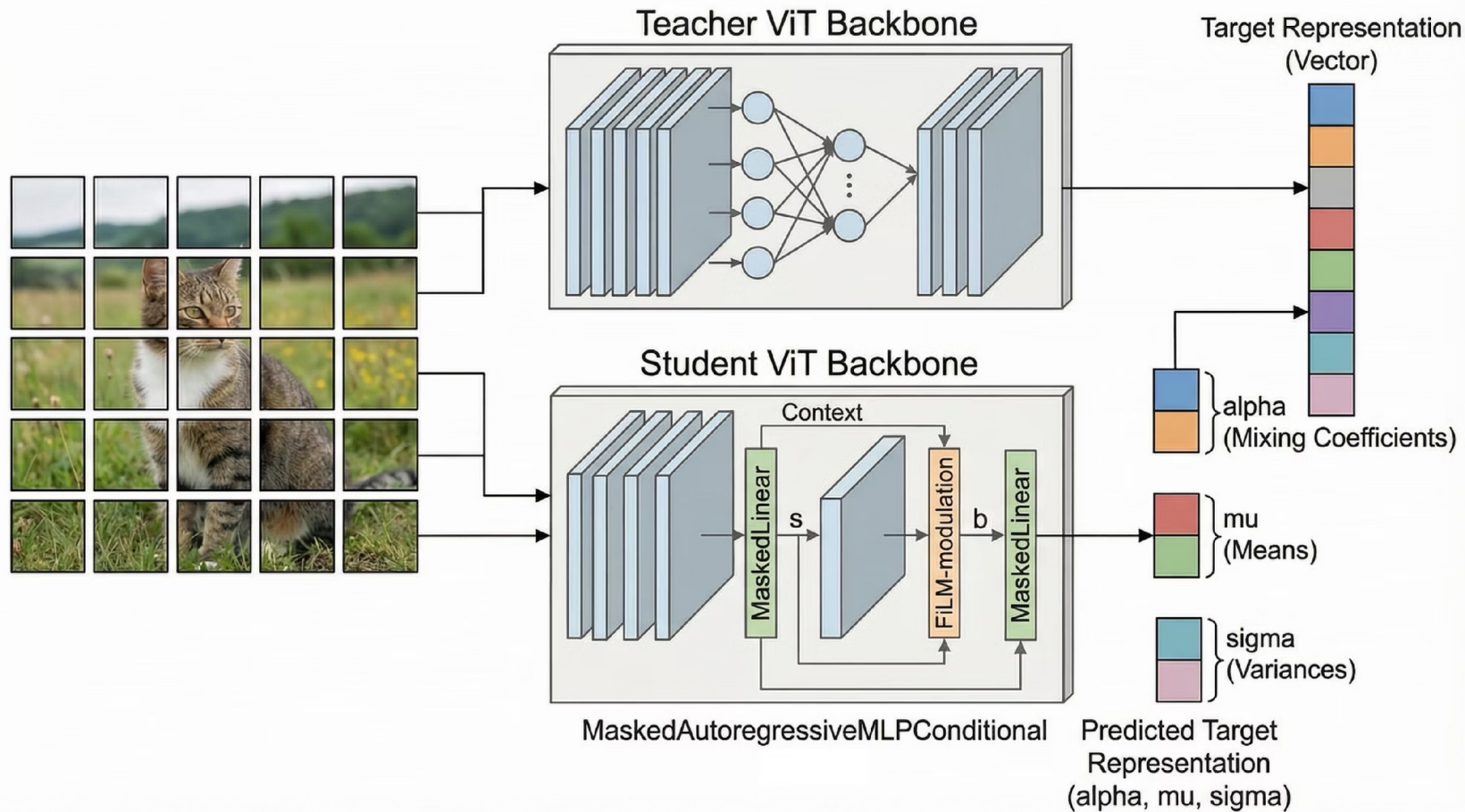
Sample dimensions sequentially

03

Apply FiLM conditioning per layer

04

Output mixture parameters coordinate-wise



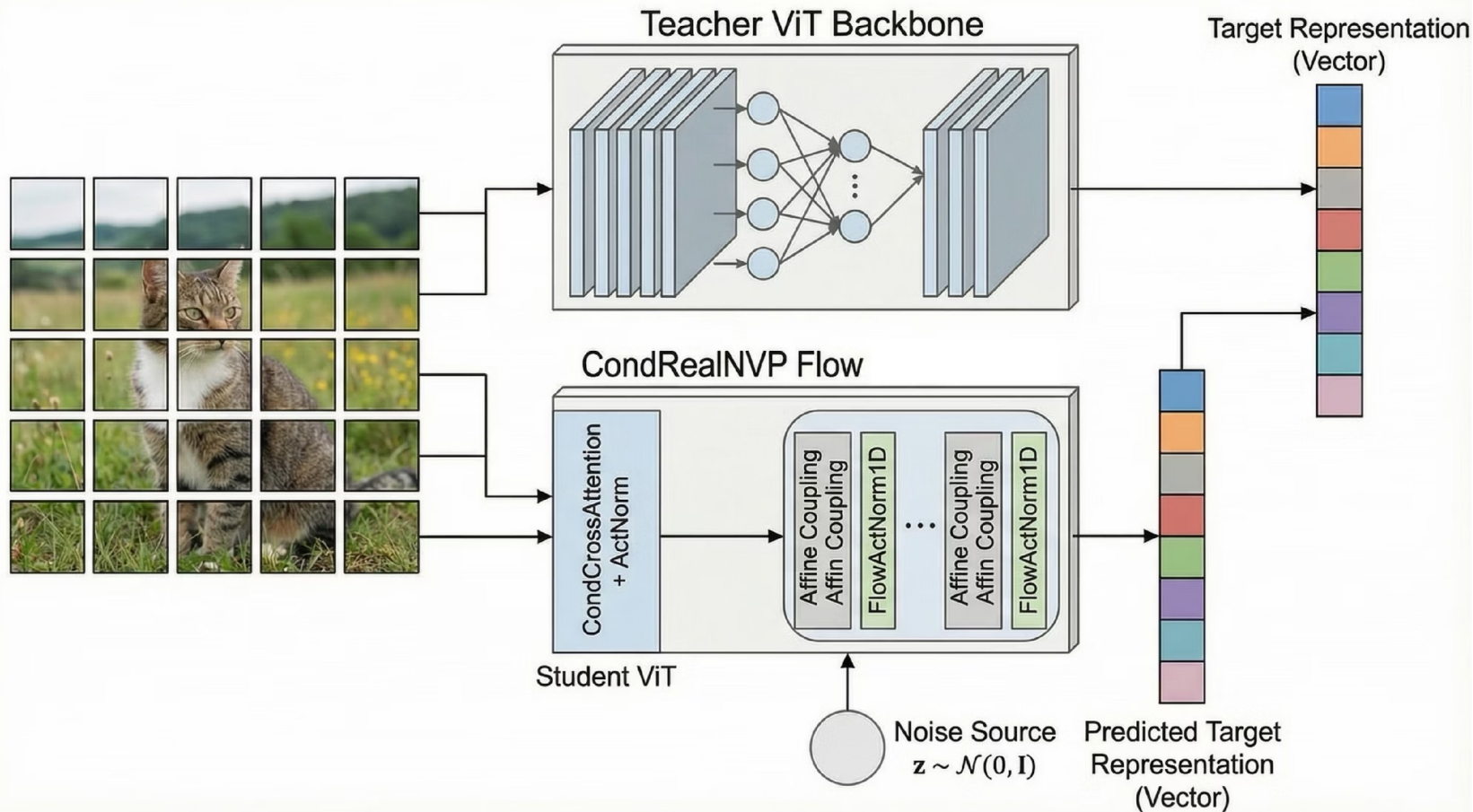
Approach 3: RealNVP Flow Matching

Key Components

- Six affine coupling layers
- ActNorm for stability
- Context + time embeddings
- Invertible transformations

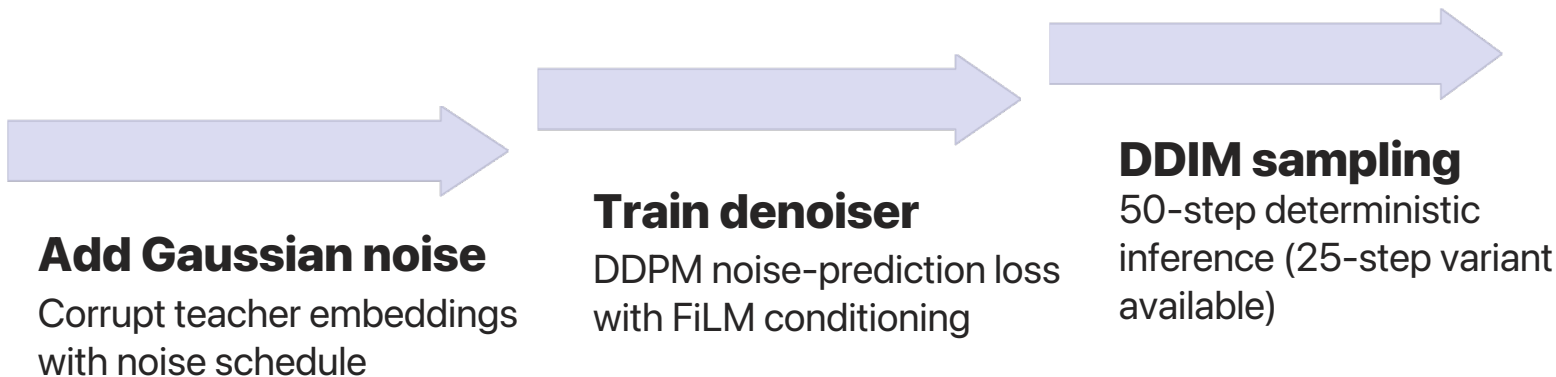
Training Strategy

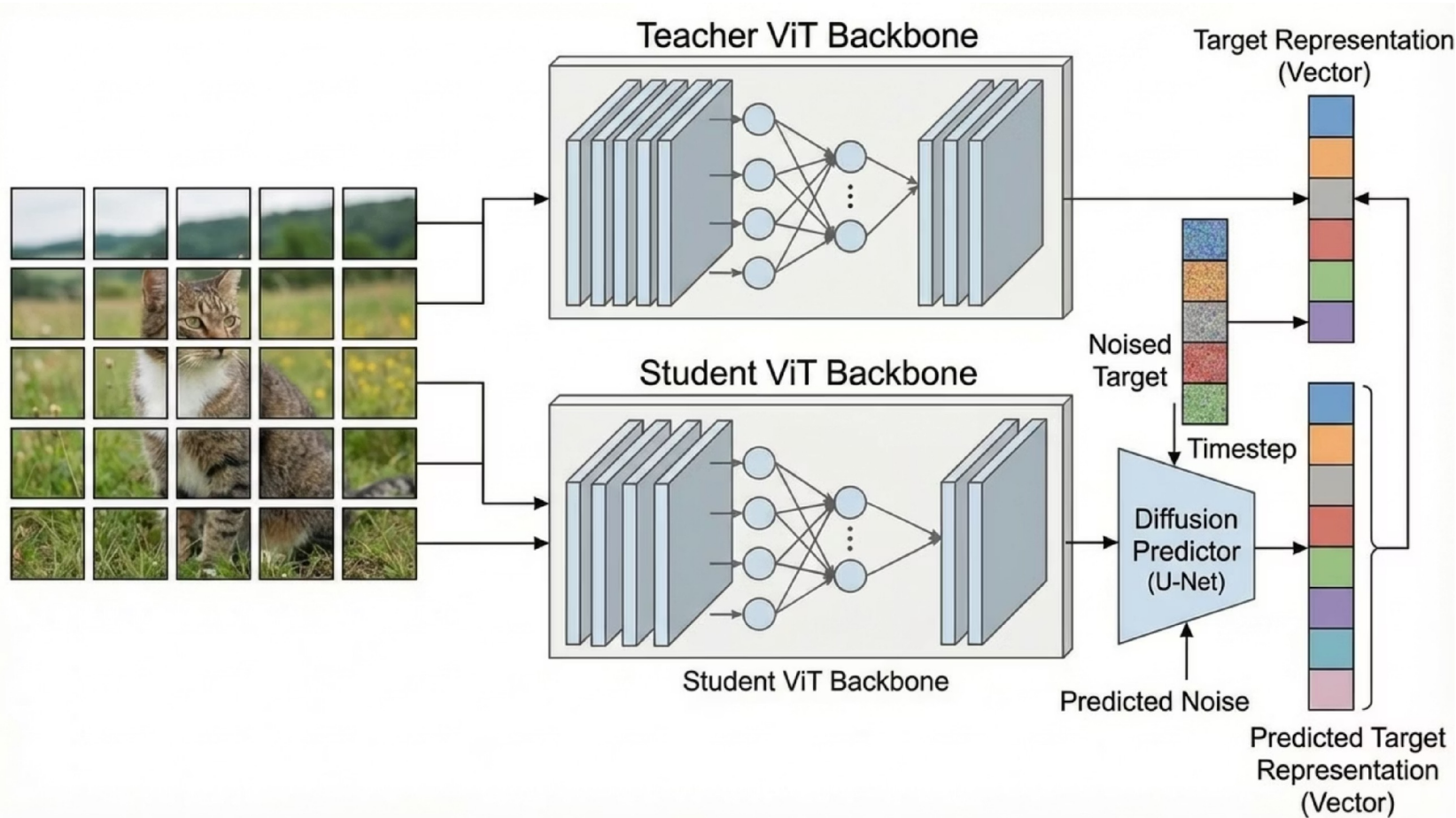
Flow matching trains with stochastic blends of teacher tokens and Gaussian noise, supervising a denoising vector field. Exact likelihood via change of variables



Approach 4: Diffusion Over Embeddings

Treats each teacher token as clean latent data. U-Net-like head denoises embeddings corrupted by linear beta schedule. Sinusoidal timestep embeddings injected into residual attention blocks





Evaluation Methodology

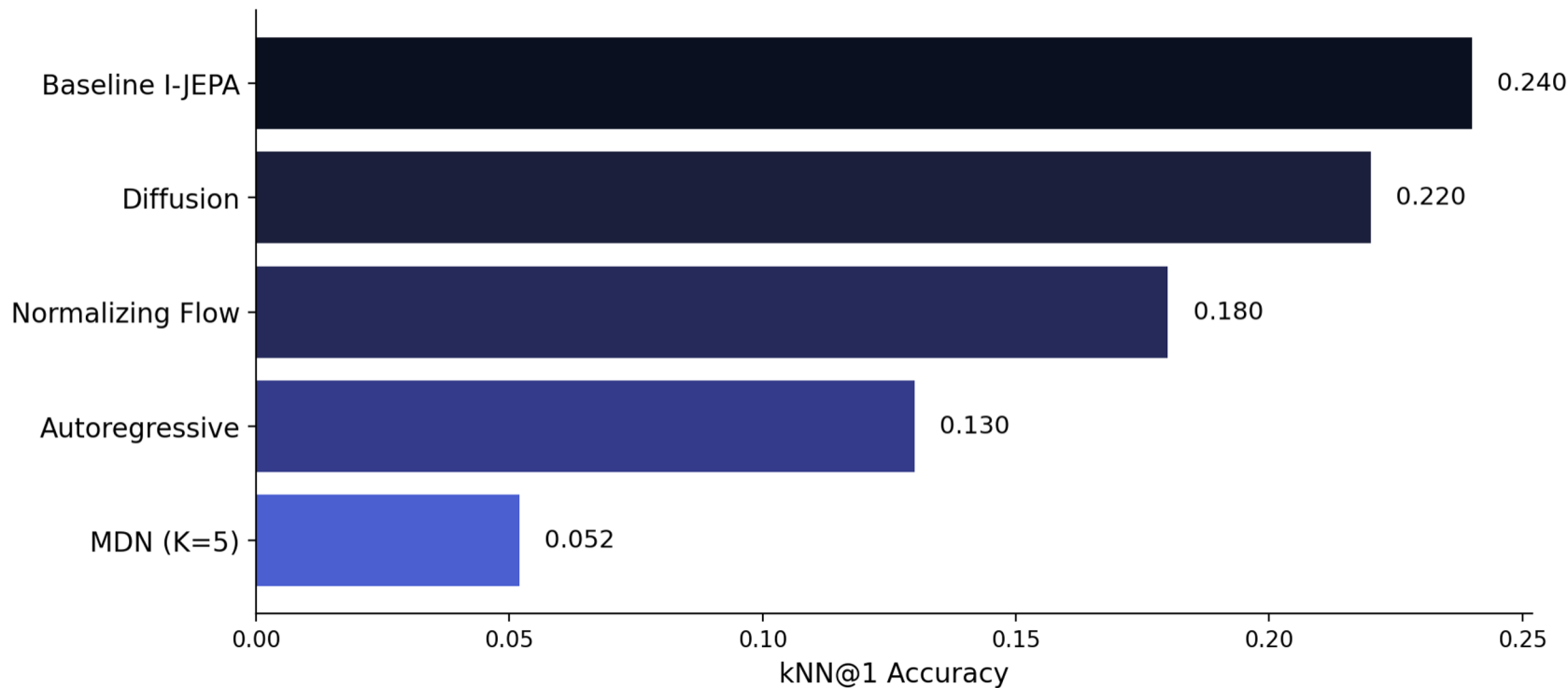
Primary Metric: kNN@1

- Freeze encoder
- Extract patch embeddings for all images
- Pool to single vector per image
- Compute k-nearest neighbors (k=20) on validation set using cosine similarity

Why kNN@1?

Measures how well embeddings cluster by semantic category and proxies how linearly separable and semantically structured features are without training a classification head

Performance Results



Training Hyperparameters

Model	Batch Size	Learning Rate	Training Steps	Epochs
Baseline	600	—	17000	—
MDN	360	3×10^{-4}	10000	75
Autoregressive	128	1×10^{-4}	20000	50
Flow	600	5×10^{-5}	15000	200
Diffusion	600	5×10^{-5}	5000	60

Analysis & Observations

Diffusion Quality

Achieves highest kNN@1 (22.36%), suggesting iterative denoising captures semantic structure most effectively

Flow Performance

Outperforms autoregressive head (18.48% vs 13.92%), likely due to exact likelihood training and richer coupling transforms

Masking Sensitivity

All variants reuse same block-sampling policy. Natural extensions include varying target area and number of blocks

EMA Dynamics

Loss curves show jagged patterns as BYOL-style EMA teacher and student chase each other, producing non-monotonic traces even as kNN improves

Future Directions

1

Baseline Alignment

Retrain deterministic I-JEPA with 384-dim architecture for apples-to-apples comparison

2

Enhanced Logging

Instrument MDN and autoregressive training with TensorBoard for real-time monitoring

3

Downstream Tasks

Add linear probes and segmentation transfers to study dense recognition improvements

4

Masking Strategies

Experiment with shared context encoders and varied masking policies for flows and diffusion

Thank you for attention!



Code Available: github.com/vladkalinichencko/Probabilistic-JEPA